



Smart Sericulture: How AI, IoT and Biotechnology Are Transforming the Silk Industry



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Abstract

Sericulture, the ancient art and science of silk production, remains a vital agro-based industry supporting millions of livelihoods, particularly in Asia. Traditional practices, however, are labour-intensive, vulnerable to environmental fluctuations, disease outbreaks and limited by biological constraints, resulting in inconsistent yields and quality. The advent of "smart sericulture" integrates cutting-edge technologies-Artificial Intelligence (AI), Internet of Things (IoT) and biotechnology-to revolutionize the sector. This comprehensive review examines how IoT-enabled real-time monitoring optimizes rearing conditions, AI-driven analytics enhance disease detection and predictive management and biotechnological interventions, including CRISPR/Cas9 and transgenic modifications, produce superior silk fibers with enhanced mechanical properties. Drawing from recent studies (2019-2025), these innovations improve productivity by up to 50%, reduce losses from diseases, enable precision farming and facilitate custom-designed silks for biomedical and industrial applications. Integrated systems combining these technologies promote sustainability, scalability and economic resilience. Despite challenges like high initial costs and technical expertise requirements, smart sericulture aligns with Industry 4.0 principles, positioning the silk industry for a technologically advanced future. This article synthesizes evidence demonstrating transformative potential while highlighting pathways for widespread adoption.

Keywords: smart sericulture, precision sericulture, artificial intelligence in sericulture, biotechnology silk production, genetic engineering silkworms and CRISPR/Cas9

Introduction

Silk production, dating back to 3500 BCE in China, has evolved from a royal luxury to a global industry valued at over \$15 billion annually. The domesticated silkworm *Bombyx mori* produces the majority of commercial silk through a multi-stage process: mulberry cultivation, egg hatching, larval rearing, cocoon spinning and reeling. Countries like China, India and Uzbekistan dominate production, with sericulture providing employment to rural populations, especially women (Fan *et al.*, 2025). Conventional sericulture faces persistent challenges: silkworms are highly sensitive to temperature (optimal 24-28°C), humidity (70-85%) and pathogens like *Beauveria bassiana* or nuclear polyhedrosis virus, causing crop losses up to 30-40%. Manual monitoring is tedious, feeding is inconsistent and genetic limitations restrict silk strength and functionality. Climate change exacerbates these issues through erratic weather and pest proliferation. "Smart sericulture" leverages digital and biological innovations to address these pain points. IoT sensors enable continuous environmental tracking, AI algorithms predict and mitigate risks and biotechnology engineers' resilient breeds with novel properties. These technologies converge under precision agriculture frameworks, often termed Sericulture 4.0 or 5.0, promising higher yields, reduced inputs and premium products (Shilpa *et al.*, 2025; Yu *et al.*, 2024). This transformation not only boosts efficiency but also opens avenues for high-value applications, such as tougher-than-Kevlar fibers or biomedical scaffolds. By reviewing recent advancements, this article illustrates how AI, IoT and biotechnology are reshaping silk farming into a modern, data-driven industry.

Methodology

This narrative review synthesizes peer-reviewed literature, conference proceedings and scholarly reports published primarily between 2019 and 2025. Sources were identified through academic databases (PubMed, ScienceDirect, ResearchGate) and web searches using terms like "AI in sericulture," "IoT sericulture monitoring," "CRISPR silkworms," "transgenic silk," and "precision sericulture." Over 50 sources were screened, with ~30 selected for depth and relevance to technological applications. Inclusion criteria prioritized empirical studies, reviews and prototypes demonstrating practical impacts on productivity, quality, or sustainability. Thematic analysis categorized content into IoT, AI, biotechnology and integrated systems. Citations adhere to APA style, with emphasis on high-impact journals (e.g., PNAS Nexus, Materials Today). No original experiments were conducted; the synthesis aims to provide an authoritative, accessible overview for researchers, farmers and policymakers.

Internet of Things (IoT) in Sericulture: Real-Time Monitoring and Automation

IoT forms the backbone of smart sericulture by deploying sensors for continuous data collection and actuators for automated responses. Environmental parameters-temperature, humidity, light and air quality-are critical for silkworm health. Traditional rearing relies on manual checks, prone to errors. IoT systems use Arduino or Raspberry Pi-based networks with DHT22 sensors, soil moisture probes and CO2 detectors to monitor rearing rooms and mulberry fields in real-time (Shilpa *et al.*, 2025). Data streams to cloud platforms (e.g., ThingSpeak or Oracle) for remote access via mobile apps. Automated controls adjust heaters, humidifiers, or ventilators, maintaining optimal conditions and reducing energy waste by 20-30%. For instance, systems trigger alerts or disinfection (UV lamps, ozone generators) upon detecting anomalies. In mulberry cultivation, IoT-enabled drip irrigation and fertigation optimize water use in drought-prone areas. Drone-integrated IoT surveys leaf health, guiding precise pesticide application. Advanced prototypes incorporate automated feeding mechanisms-conveyor belts or robotic arms dispensing chopped mulberry leaves-minimizing labor and ensuring uniform nutrition. Cocoon harvesting robots further streamline processes. Case studies from India and China report 15-25% yield improvements and 40% labor reduction with IoT adoption. These systems enhance traceability, supporting premium pricing for tech-certified silk (Various IoT papers, 2024-2025).

Artificial Intelligence Applications: Predictive Analytics and Disease Management

AI elevates IoT data into actionable insights through machine learning (ML) and deep learning. Disease detection is a prime application. Silkworm pathologies (e.g., flacherie, grasserie) manifest visually or behaviourally. Convolutional Neural Networks (CNNs) analyse images from IoT cameras, achieving >95% accuracy in identifying infections early-far superior to human observation (Shilpa *et al.*, 2025). Predictive models forecast outbreaks using historical data, weather APIs and sensor inputs. Time-series algorithms (LSTM) predict yield with <10% error, aiding planning. Precision feeding optimizes leaf allocation based on larval stage and growth rates, monitored via weight sensors or computer vision. Expert systems guide farmers on best practices, while generative AI simulates scenarios for training. In marketing, AI analyses demand patterns for silk varieties. Integrated with blockchain, AI ensures supply chain transparency. Studies show AI reduces disease losses by 50-70%, with systems like "Silk Shield" combining image processing for multi-disease detection (2025 reviews).

Biotechnology Advancements: Engineering Superior Silkworms and Silk

Biotechnology targets genetic limitations, creating resilient, high-performance breeds. CRISPR/Cas9 enables precise editing. Early applications knocked out disease-susceptibility genes; recent work inserts exogenous sequences for novel fibers. Transgenic silkworms incorporate spider dragline genes (e.g., MaSp from *Nephila clavipes*), producing tougher silk. Yu *et al.* (2024) used TALEN and piggyBac transposons to integrate large spidroin genes, yielding fibers with 86% higher toughness and 40% increased crystallinity. Site-specific integration minimizes disruption to endogenous genes, achieving high expression (up to 64% exogenous protein). Other modifications enhance disease resistance (e.g., antimicrobial peptides) or environmental tolerance. RNA interference (RNAi) silences undesirable traits. Silk gland bioreactors produce recombinant proteins for pharmaceuticals-collagen, vaccines-diversifying

revenue. Fan *et al.* (2025) review strategies: exogenous introduction, endogenous optimization, gland regulation and yield enhancement, noting mechanical improvements rival synthetic materials.

Integrated Smart Sericulture Systems: Convergence of Technologies

- True transformation occurs through synergy.
- Frameworks combine IoT sensors with AI analytics and biotechnological breeds. For example, transgenic disease-resistant silkworms in IoT-monitored rooms, with AI predicting optimal conditions.
- Precision sericulture platforms use drones for field mapping, big data for decision support and automated rearing units.
- Chinese facilities employ fully digitized lines for mass cocoon production, integrating robotics and AI (2024 reports).
- Blockchain traces from genetically verified eggs to final silk, appealing to ethical markets.
- These systems boost sustainability: reduced chemicals via precise interventions, lower water via optimized irrigation

Economic and Socio-Economic Impacts

- Smart sericulture increases yields 20-50%, cuts costs 30% and enables premium products (e.g., spider-silk hybrids).
- Rural empowerment through apps democratizes knowledge; women benefit from reduced physical labor.
- Global competitiveness rises, with custom silks for aerospace or medicine.

Challenges and Limitations

- High upfront costs limit smallholder adoption; digital literacy and infrastructure gaps persist in rural areas.
- Data privacy, algorithmic bias and biosafety concerns (transgenics) require regulation.
- Scalability of biotech demands streamlined approvals.

Future Prospects

Sericulture 5.0 envisions fully autonomous farms with swarm robotics, advanced AI (e.g., reinforcement learning) and multi-gene edits for multifunctional silks. Nanotechnology supplements (e.g., fortified diets) and climate-resilient breeds address global warming. Policy support-subsidies, training-will accelerate diffusion.

Conclusion

Smart sericulture exemplifies how AI, IoT and biotechnology converge to modernize a millennia-old industry. From real-time IoT monitoring to AI-driven predictions and genetically engineered super-silks, these innovations enhance efficiency, resilience and value. Backed by robust evidence (Yu *et al.*, 2024; Shilpa *et al.*, 2025; Fan *et al.*, 2025), the sector is transitioning toward precision, sustainability and innovation. Overcoming barriers through collaboration will ensure sericulture thrives in the digital age, delivering luxurious, ethical silk while supporting livelihoods and scientific progress.

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